

Step by step implementation of Custom Amazon Machine Image (AMI) using standard Ubuntu Linux EC2 instance with R programming language.

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Question 2 – Task A – Part 1.

Step 1 – Sign in to AWS and Navigate to EC2 Dashboard.

- **Sign in** to the **AWS Management Console**, find “**Services**” in the top menu (left side), and directly search and click “**EC2**” as shown in figure 1.

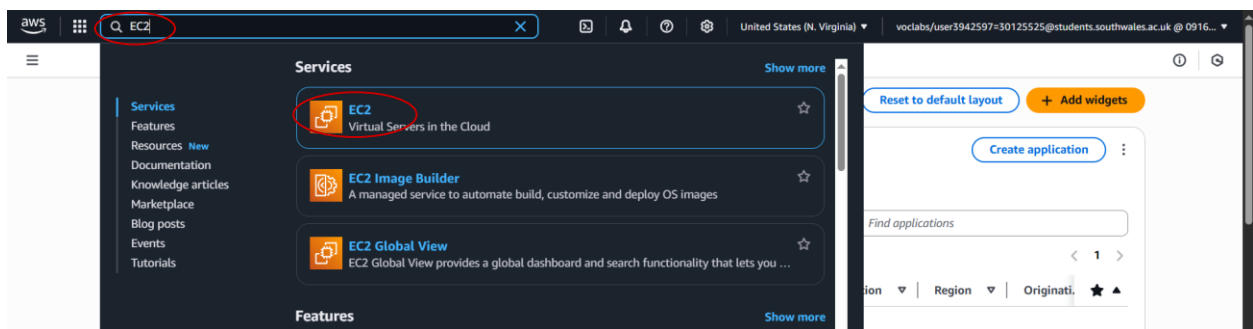


Figure 1. Navigating to EC2 Dashboard.

Step 2 – Launch a New Instance.

- Click “**Instances**” which is on the left panel and then click “**Launch Instances**” as depicted in figure 2.

This is to start the process of creating a new virtual machine.

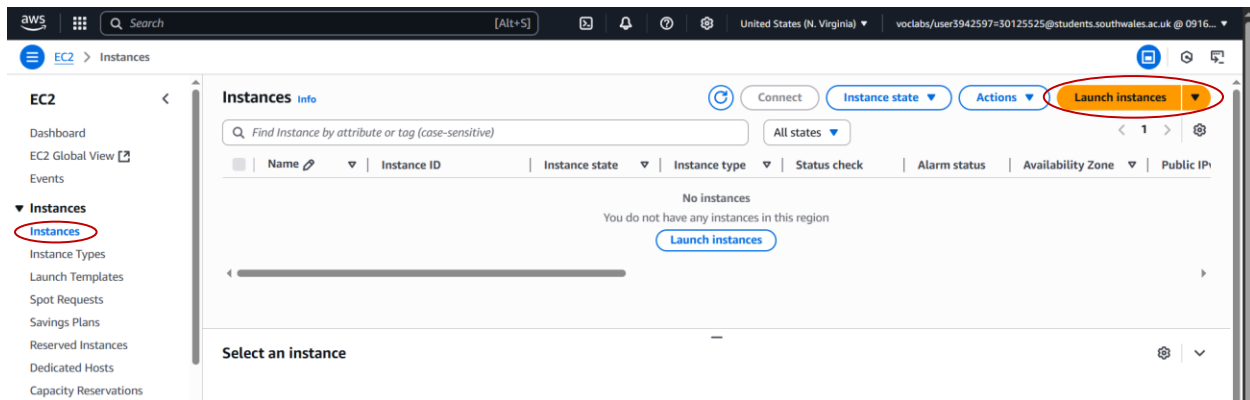


Figure 2. Launching an EC2 Instance.

Step 3 – Naming the instance.

- Under “**Name and tags**” give the desired name of your choice. Here, we will name our instance as “**30125525-ec2-template-instance**” as shown in figure 3.
- The name should be unique for traceability.

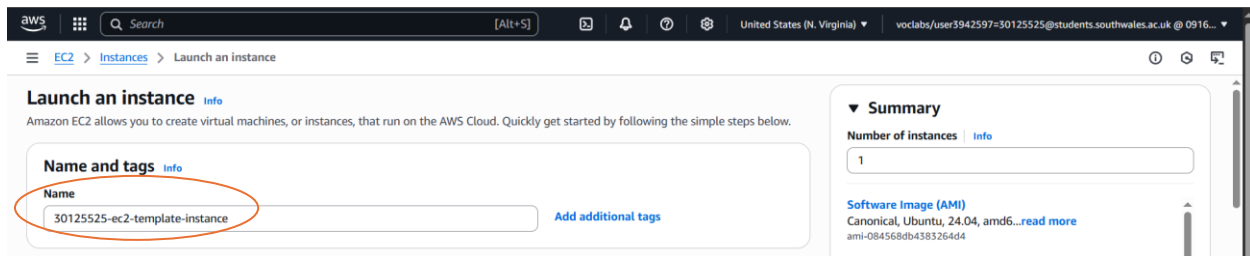


Figure 3. Naming an EC2 Instance.

Step 4 – Choosing AMI.

- Under “**Application and OS Images (Amazon Machine Image)**” select “**Ubuntu 24.04 LTS**” which is a free tier with a recent publish date as shown in figure 4.

Ubuntu is a stable environment for R Programming.

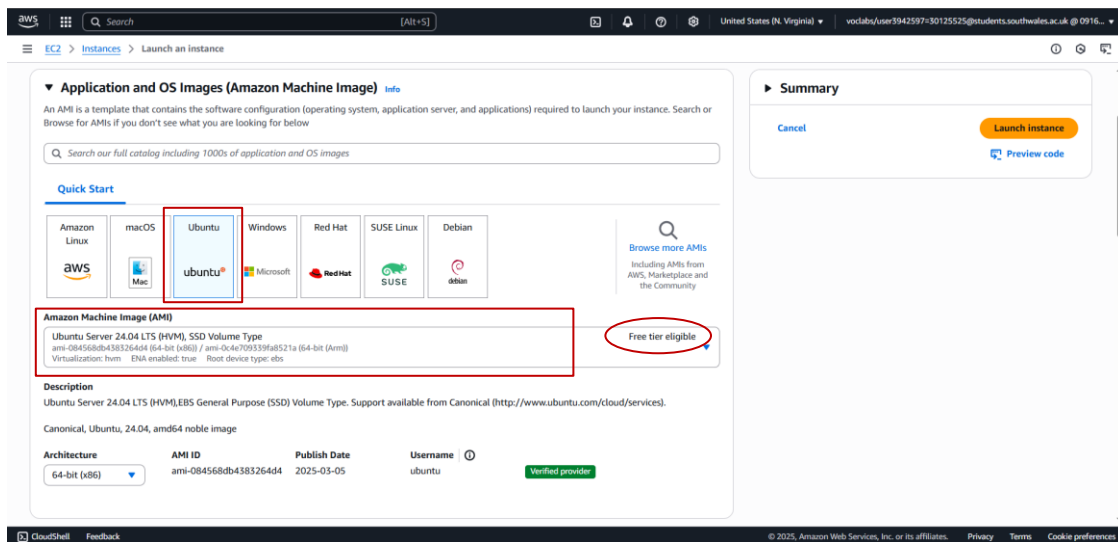


Figure 4. Selecting Ubuntu for AMI.

Step 5 – Choosing Instance Type.

- Under “**Instance Type**” select the best suited instance based on your budget and workload. Here, we will select “**t2.nano**” as shown below in figure 5.

We have opted for “t2.nano” since it is a low-cost, free tier eligible instance.

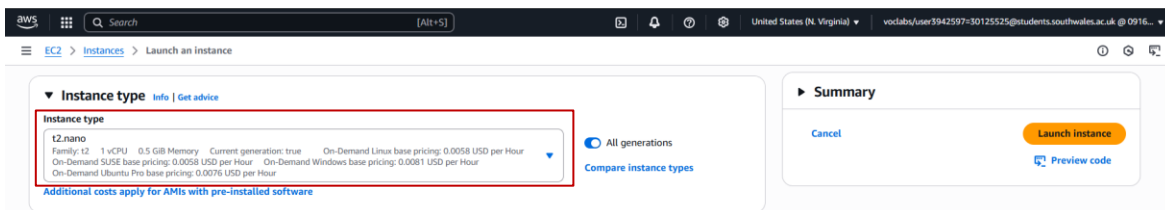


Figure 5. Choosing Instance Type as “t2.nano”.

Step 6 – Creating a Key Pair (Login).

- Under “**Key pair (login)**” click “**Create new key pair**” to create a key pair as shown in figure 6.

It is important to have a key pair so that we can use SSH access to the instance.

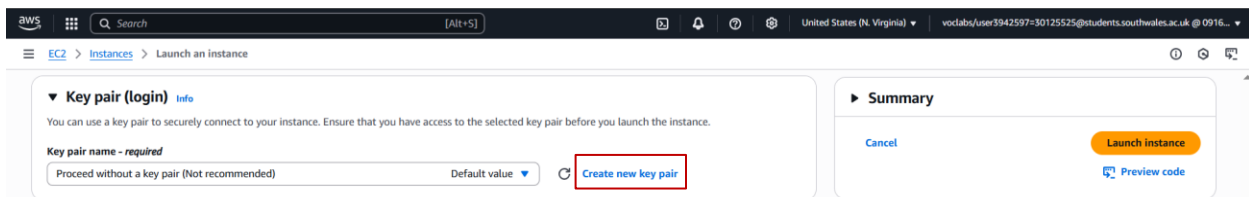


Figure 6. Creating a new key pair.

- Next, type in the desired key name. Here, we are naming our key pair as “**30125525-NewKey**”. Then, under “**Key pair type**” select “**RSA**”.
- For Private key file format select “**.pem**” and finally click “**Create key pair**” to create it as marked in figure 7.

Create key pair

Key pair name
Key pairs allow you to connect to your instance securely.
30125525-NewKey
The name can include up to 255 ASCII characters. It can't include leading or trailing spaces.

Key pair type
☒ **RSA**
RSA encrypted private and public key pair
☐ **ED25519**
ED25519 encrypted private and public key pair

Private key file format
☒ **.pem**
For use with OpenSSH
☐ **.ppk**
For use with PuTTY

⚠ When prompted, store the private key in a secure and accessible location on your computer. You will need it later to connect to your instance. [Learn more](#)

[Cancel](#) [Create key pair](#)

Figure 7. Finalizing key pair name and properties.

- After completing the above steps, a “**30125525-NewKey.pem**” file will be downloaded. Keep it securely as it will be needed later.

Step 7 – Configuring Network Setting.

- Under “**Network Settings**” ensure that “**Auto-assign public IP**” is “**Enabled**” as shown in figure 8.
- Tick “**Allow SSH traffic from**” and IP as “**Anywhere**” for simplicity as shown in figure 8.

Network settings [Info](#)

Network [Info](#)
vpc-0914a15bc907b5c49

Subnet [Info](#)
No preference (Default subnet in any availability zone)

Auto-assign public IP [Info](#)
Enable

Additional charges apply when outside of free tier allowance

Firewall (security groups) [Info](#)
A security group is a set of firewall rules that control the traffic for your instance. Add rules to allow specific traffic to reach your instance.

☒ **Create security group** ☐ Select existing security group

We'll create a new security group called 'launch-wizard-1' with the following rules:

☒ **Allow SSH traffic from**
Helps you connect to your instance. **Anywhere** (0.0.0.0/0)

☐ Allow HTTPS traffic from the internet
To set up an endpoint, for example when creating a web server

☐ Allow HTTP traffic from the internet
To set up an endpoint, for example when creating a web server

[Edit](#)

Summary

[Cancel](#) [Launch instance](#) [Preview code](#)

Figure 8. Setting up Network Settings.

Step 8 – Configuring Storage.

- Under “Configure storage” set the storage to “8 GiB” of “gp3” storage as shown in figure 9.

This is to enable general-purpose SSD volume type.

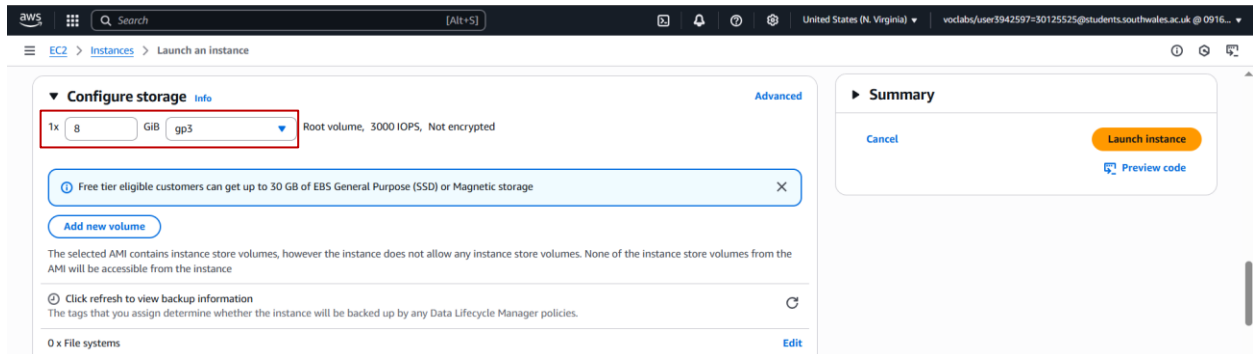


Figure 9. Setting up Storage.

Step 9. Enable Termination Protection.

- Under “Advanced details” enable “Termination Protection” as shown in below in figure 10.

This will be useful to prevent accidental termination of the instance.

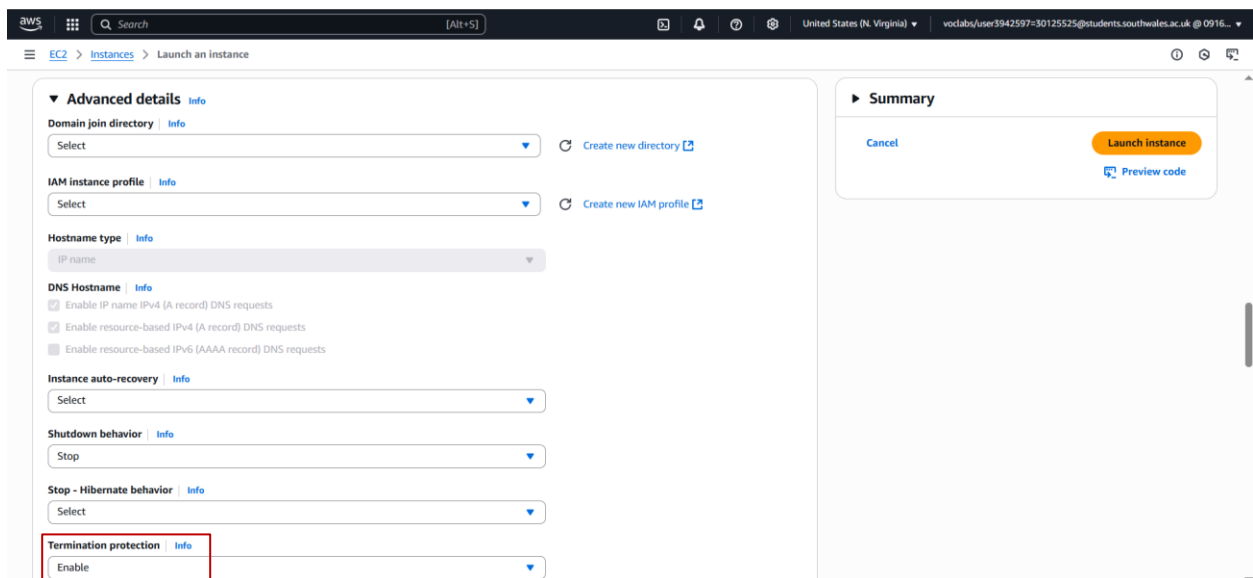


Figure 10. Enabling Termination Protection.

Step 10 – Launch the Instance.

- After reviewing all the checkpoints, click “**Launch Instance**” as shown in figure 11.
- After these steps we will launch our first instance.

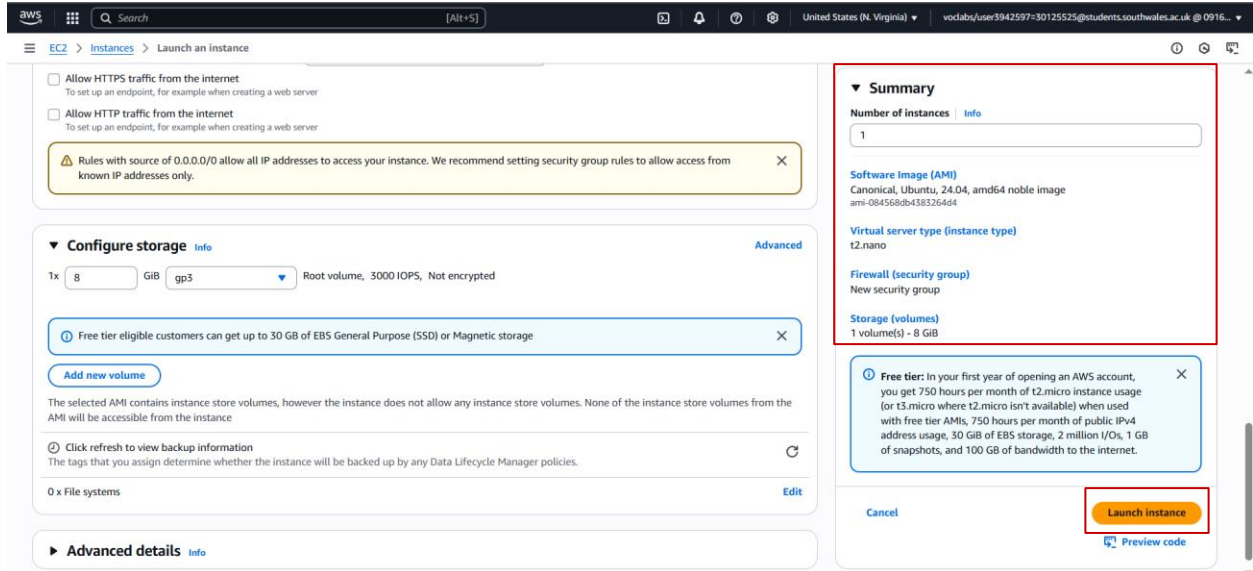


Figure 11. Launching our first Instance.

- Under “**Instances**” we can find our newly created instance running with all checks cleared as in figure 12.

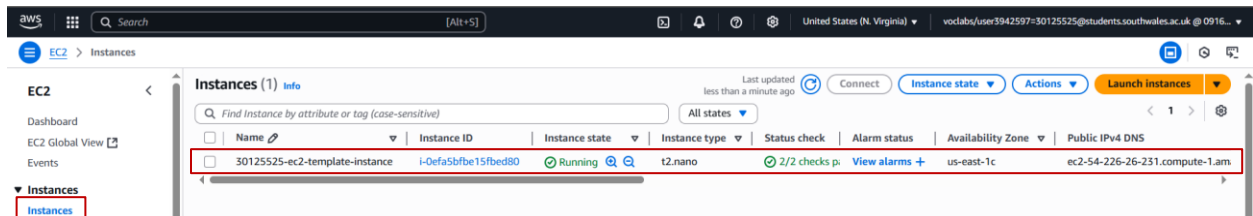


Figure 12. Newly created Instance.

Step 11 – Connecting the Instance.

- Click “**Connect**” as shown in figure 13 and click “**Connect**” to open bash terminal as shown in figure 14.

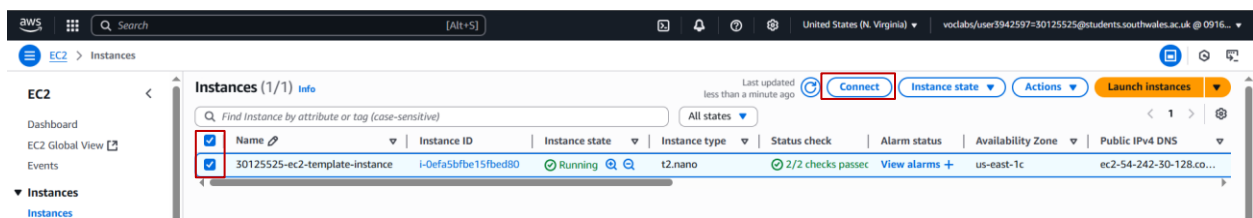


Figure 13. Connect EC2 Instance

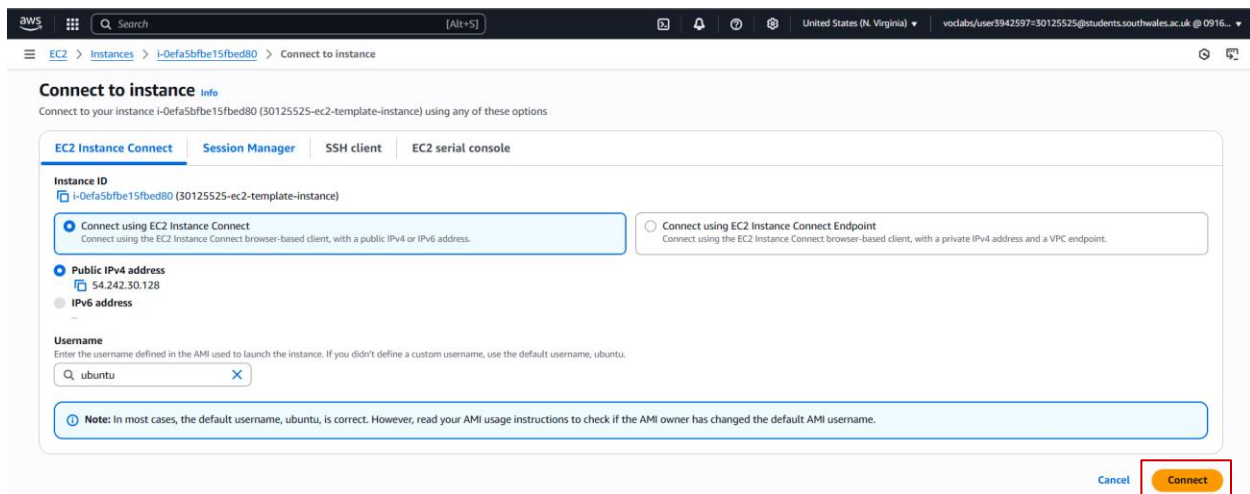


Figure 14. Click Connect to open the bash terminal.

Step 12 – Installing R Language.

- After clicking “**Connect**” a new page of bash terminal will open.
- Run “**sudo apt update**” to update the packages as shown in figure 15.
- Run “**sudo apt install -y r-base**” to install R Language as shown in figure 16.
- To verify if it is installed or not, run “**R --version**” as shown in figure 17.
- Finally, type “**Exit**” to logout.

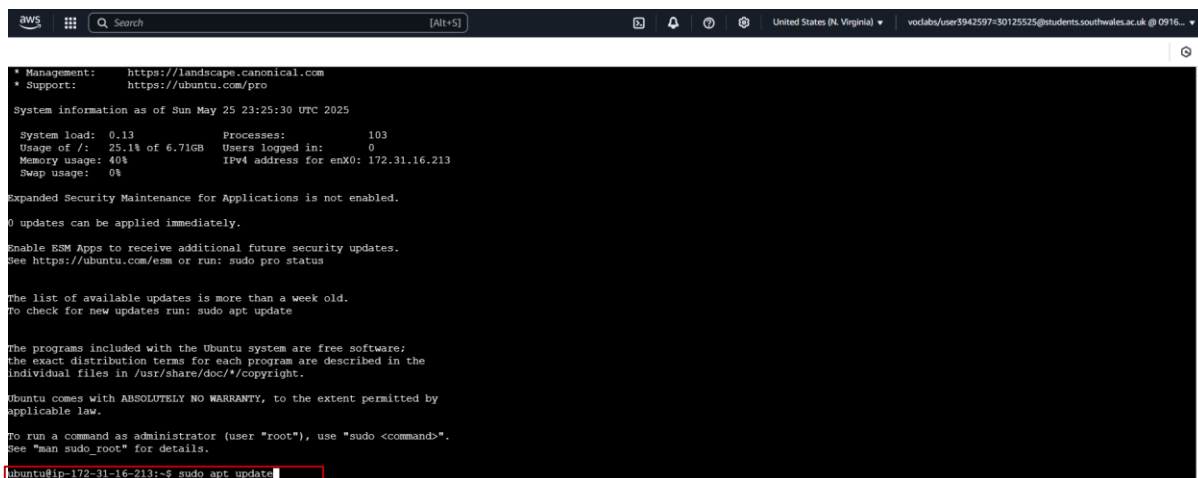
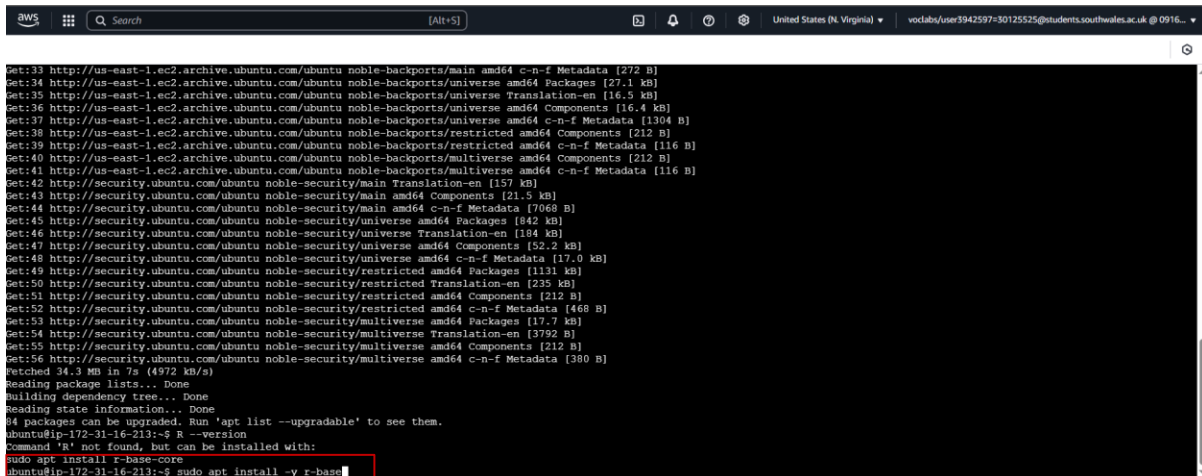
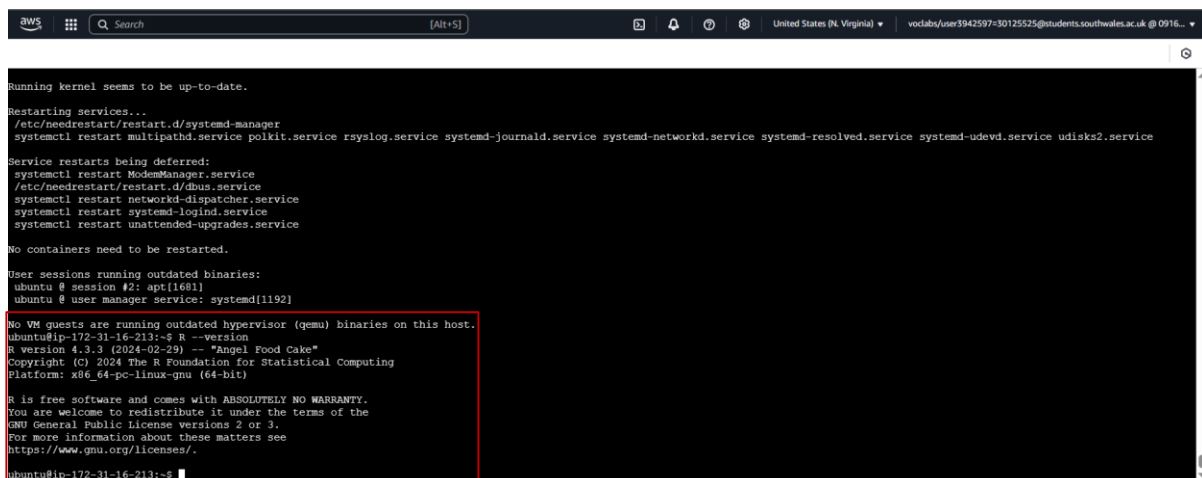


Figure 15. Updating to the latest packages.



```
Set:33 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/main amd64 c-n-f Metadata [272 B]
Set:34 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Packages [27.1 kB]
Set:35 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe Translation-en [16.5 kB]
Set:36 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 Components [16.4 kB]
Set:37 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/universe amd64 c-n-f Metadata [1304 B]
Set:38 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 Components [212 B]
Set:39 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/restricted amd64 c-n-f Metadata [116 B]
Set:40 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 Components [212 B]
Set:41 http://us-east-1.ec2.archive.ubuntu.com/ubuntu noble-backports/multiverse amd64 c-n-f Metadata [116 B]
Set:42 http://security.ubuntu.com/ubuntu noble-security/main amd64 Translation-en [157 kB]
Set:43 http://security.ubuntu.com/ubuntu noble-security/main amd64 Components [21.5 kB]
Set:44 http://security.ubuntu.com/ubuntu noble-security/main amd64 c-n-f Metadata [7068 B]
Set:45 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Packages [842 kB]
Set:46 http://security.ubuntu.com/ubuntu noble-security/universe Translation-en [184 kB]
Set:47 http://security.ubuntu.com/ubuntu noble-security/universe amd64 Components [52.2 kB]
Set:48 http://security.ubuntu.com/ubuntu noble-security/universe amd64 c-n-f Metadata [17.0 kB]
Set:49 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Packages [1131 kB]
Set:50 http://security.ubuntu.com/ubuntu noble-security/restricted Translation-en [235 kB]
Set:51 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 Components [212 B]
Set:52 http://security.ubuntu.com/ubuntu noble-security/restricted amd64 c-n-f Metadata [468 B]
Set:53 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Packages [17.7 kB]
Set:54 http://security.ubuntu.com/ubuntu noble-security/multiverse Translation-en [3792 B]
Set:55 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 Components [212 B]
Set:56 http://security.ubuntu.com/ubuntu noble-security/multiverse amd64 c-n-f Metadata [380 B]
Fetched 34.3 MB in 7s (4972 kB/s)
Reading package lists... Done
Building dependency tree... Done
Reading state information... Done
14 packages can be upgraded. Run 'apt list --upgradable' to see them.
ubuntu@ip-172-31-16-213:~$ R --version
Command 'R' not found, but can be installed with:
sudo apt install r-base-core
ubuntu@ip-172-31-16-213:~$ sudo apt install -y r-base
```

Figure 16. Installing R package.



```
Running kernel seems to be up-to-date.
Restarting services...
/etc/needrestart/restart.d/systemd-manager
systemctl restart multipathd.service polkit.service rsyslog.service systemd-journald.service systemd-networkd.service systemd-resolved.service systemd-udev.service udisks2.service

Service restarts being deferred:
systemctl restart ModemManager.service
/etc/needrestart/restart.d/dbus.service
systemctl restart networkd-dispatcher.service
systemctl restart systemd-logind.service
systemctl restart unattended-upgrades.service

No containers need to be restarted.

User sessions running outdated binaries:
ubuntu # session #2: apt[168]
ubuntu # user manager service: systemd[1192]

No VM guests are running outdated hypervisor (qemu) binaries on this host.
ubuntu@ip-172-31-16-213:~$ R --version
R version 4.3.3 (2024-02-29) -- "Angel Food Cake"
Copyright (C) 2024 The R Foundation for Statistical Computing
Platform: x86_64-pc-linux-gnu (64-bit)

R is free software and comes with ABSOLUTELY NO WARRANTY.
You are welcome to redistribute it under the terms of the
GNU General Public License versions 2 or 3.
For more information about these matters see
https://www.gnu.org/licenses/.
ubuntu@ip-172-31-16-213:~$
```

Figure 17. Verifying installation of R package.

Step 13 – Stopping Instance to create a Custom AMI.

- First, we must stop the running instance to create a custom AMI. Select your created instance and under “**Instance state**” click “**Stop instance**” as shown in figure 18.

It helps to keep consistency and prevents data corruption.

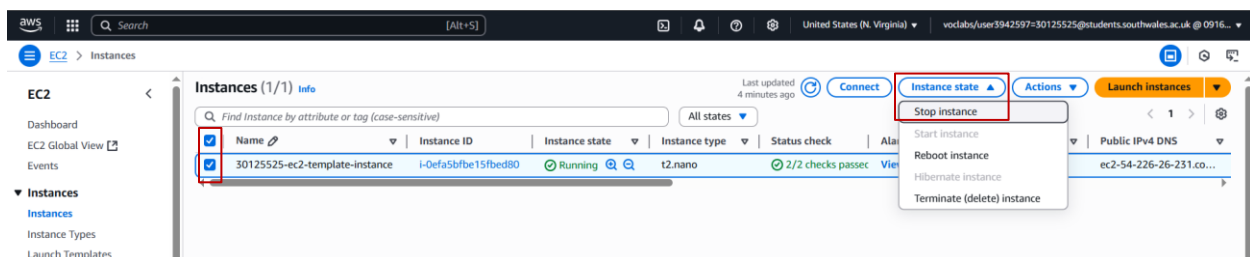


Figure 18. Stopping the running instance.

Step 14 – Creating a Custom AMI.

- Select the instance and under “**Actions**” then go to “**Image and Templates**” and click “**Create Image**” as depicted in figure 19.

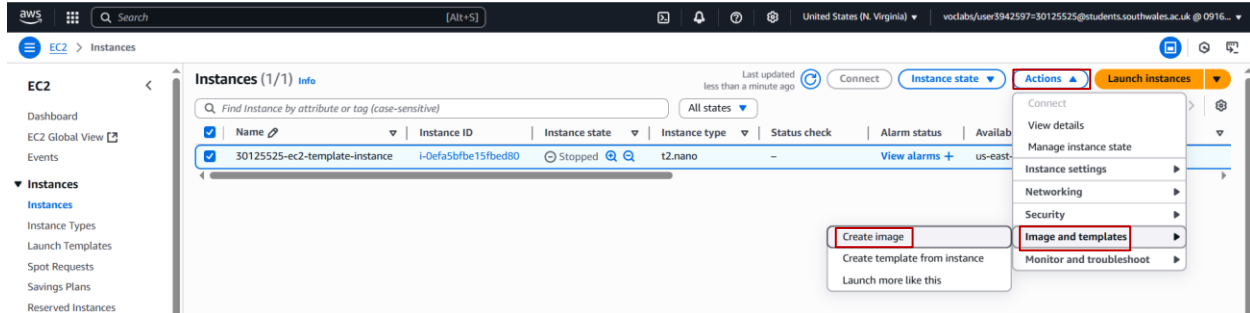


Figure 19. Creating an AMI Image.

Step 15 – Finalizing Custom AMI.

- Under “**Image name**” write the desired name you want to. Here, we are naming “**30125525-r-instance**” and a small description as shown in figure 20.

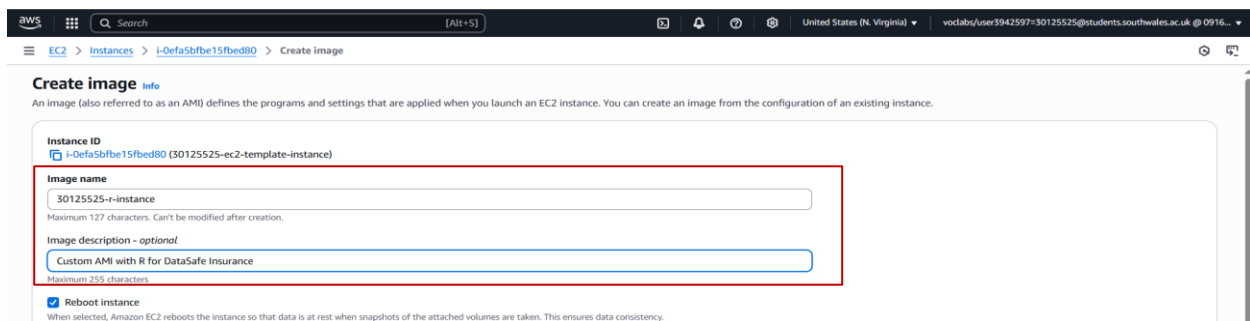


Figure 20. Naming Image.

- Finally, click “**Create image**” at the bottom right of the page to create the custom AMI named “**30125525-r-instance**”.
- Click “**AMIs**” which is on the left panel to check our newly created AMI as shown in figure 21.

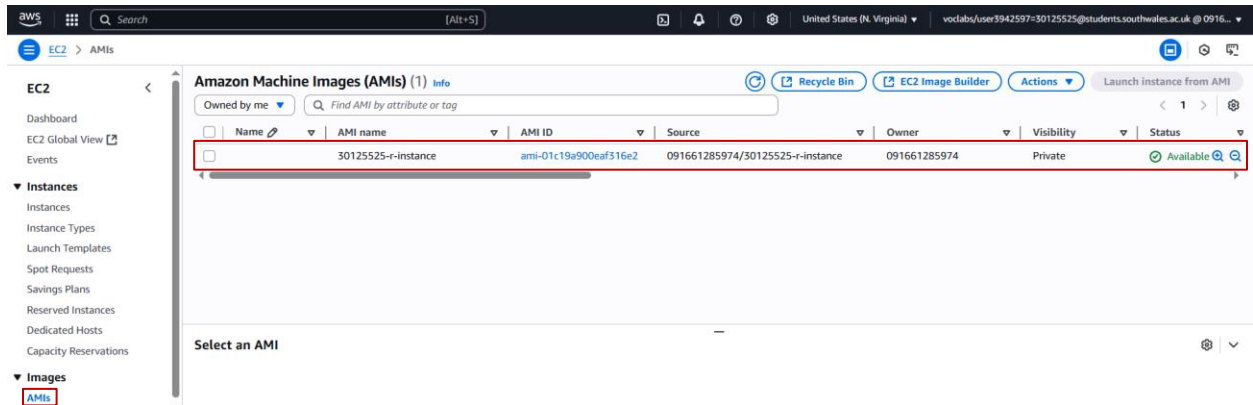


Figure 21. Custom AMI with pre-installed R Programming Language.

Question 2 – Task A – Part 2.

Step 1 – Sign in to AWS and Navigate to EC2 Dashboard.

- **Sign in to the AWS Management Console**, find “**Services**” in the top menu (left side), and directly search and click “**EC2**” as shown in figure 1.

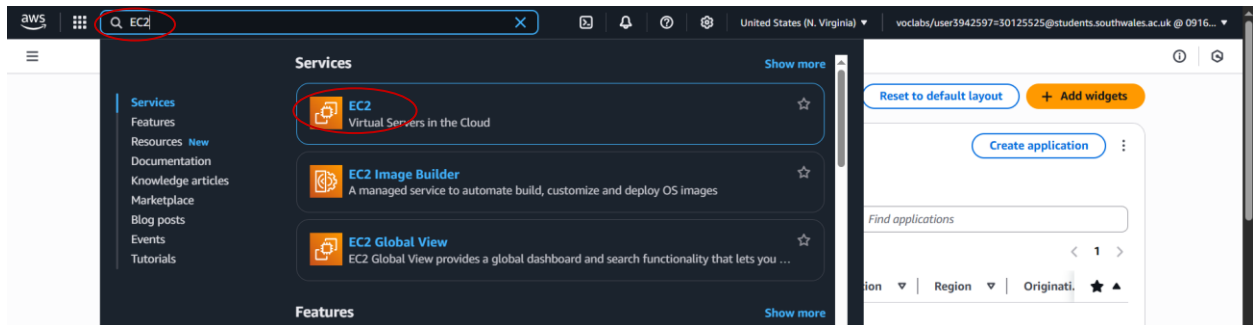


Figure 1. Navigating to EC2 Dashboard.

Step 2 – Navigating the newly created custom AMI.

- Click “**AMIs**” which is on the left panel to navigate to our newly created AMI with pre-installed R Programming Language as depicted in figure 2.

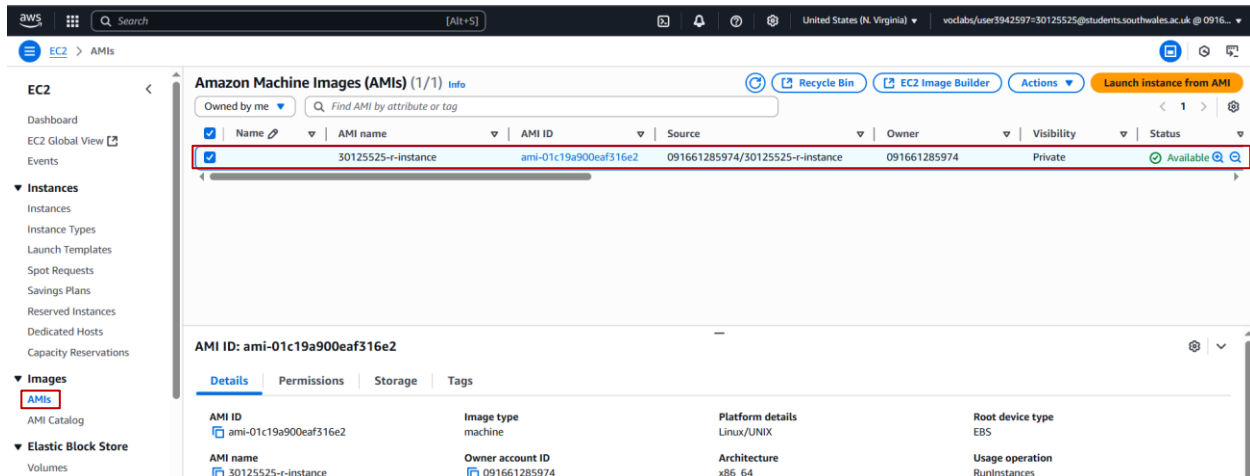


Figure 2. Navigating to Custom AMI.

Step 3 – Launching Test Instance.

- Click **“Launch instance from AMI”** to an instance from our custom AMI as shown in figure 3.

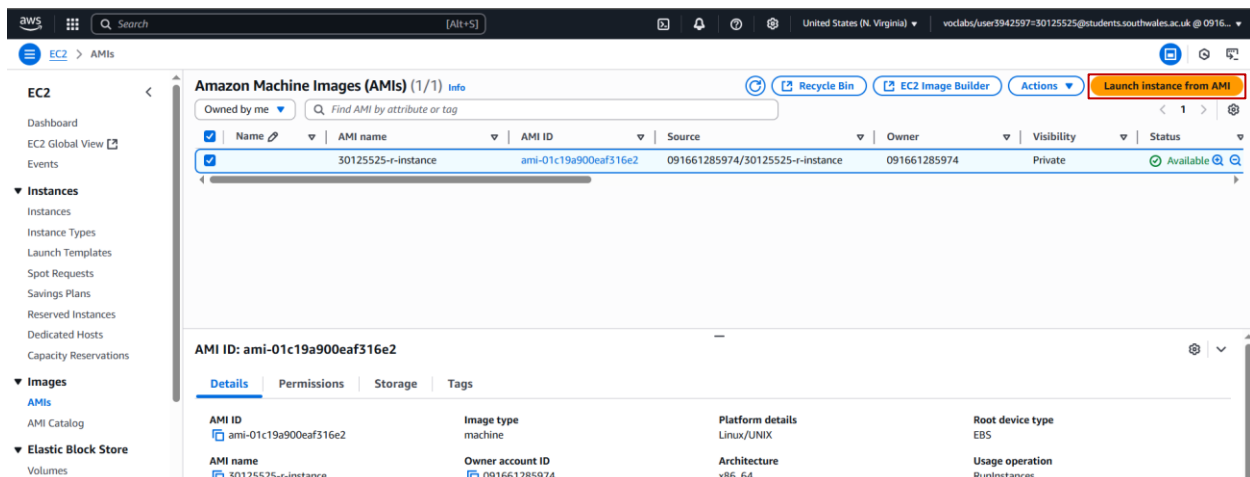


Figure 3. Launching Instance from our Custom AMI.

Step 4 – Customizing the Test Instance.

- You can give any desired name as per your naming convention. Here, we will name the instance as **“30125525-datasafe-r-instance”** as shown below in figure 4.
- Under **“Application and OS Images (AMI)”** our custom AMI is selected by default.

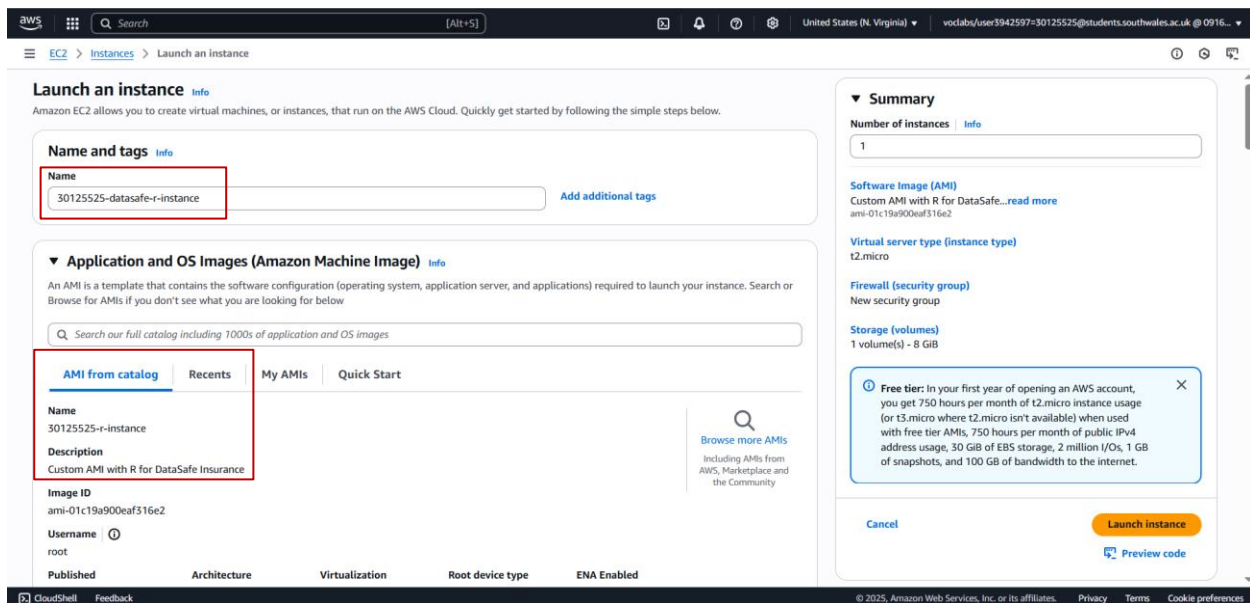


Figure 4. Naming the Test Instance.

Step 5 – Update Instance Type.

- Under “**Instance Type**” select “**t2.nano**” as highlighted in figure 5.

We have opted for “t2.nano” since it is a low-cost, free tier eligible instance. Select the best suited instance based on your budget and workload.

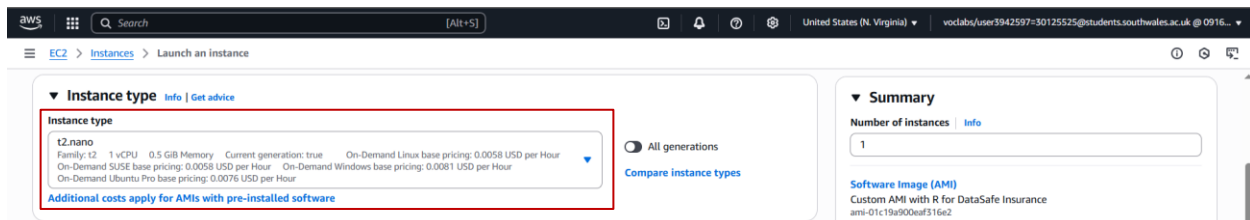


Figure 5. Selecting Instance Type.

Step 6 – Choosing Key Pair (login).

- Under “**Key pair name**”, select the recently created key “**30125525-NewKey**” as highlighted in figure 6.

It is important to have a key pair so that we can use SSH access to the instance.

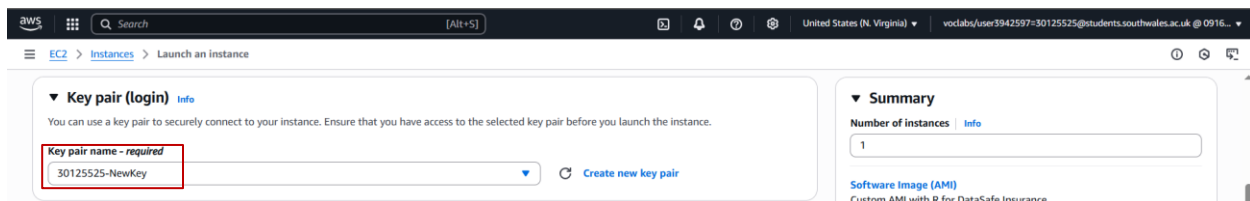


Figure 6. Selecting Key Pair (Login).

Step 7 – Configuring Configure Storage.

- Under “Configure storage” set the storage to “8 GiB” of “gp3” storage as shown in figure 7.

This is to enable general-purpose SSD volume type.

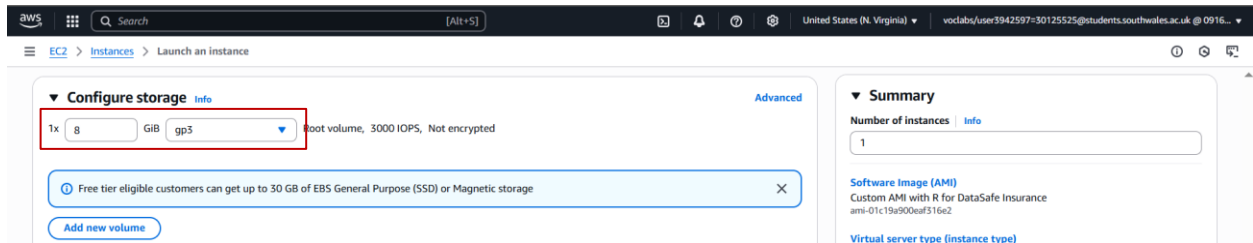


Figure 7. Customizing Storage.

Step 8 – Enabling Termination Protection.

- Under “Advanced details” enable “Termination Protection” as shown in below in figure 8.

This will be useful to prevent accidental termination of the instance.

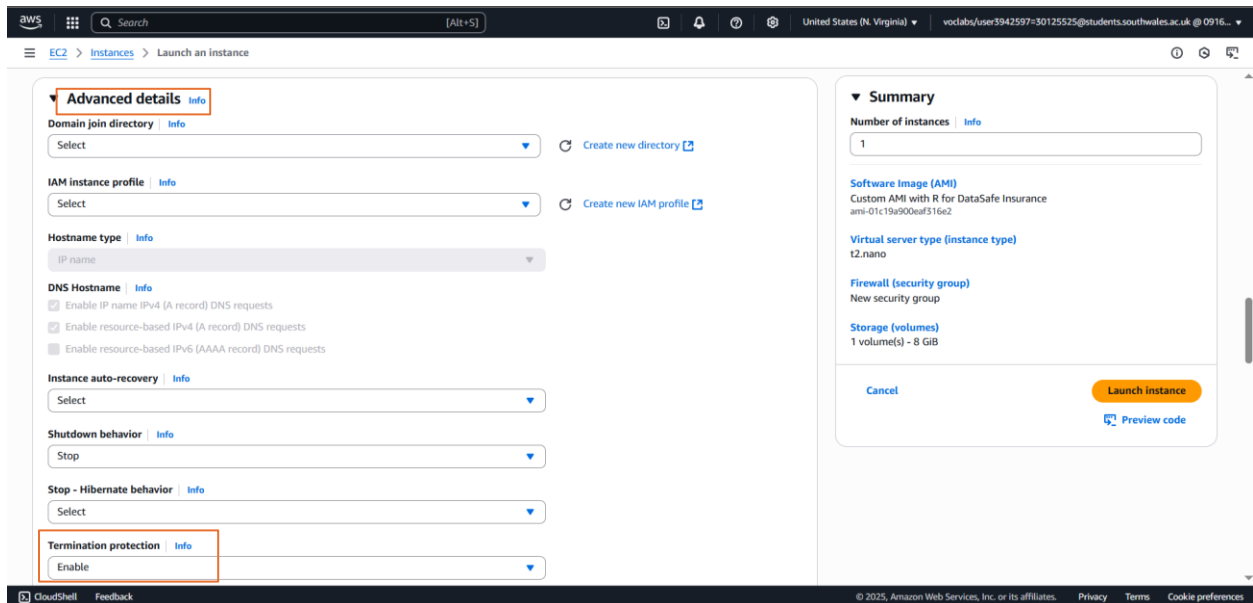


Figure 8. Enabling Termination Protection.

Step 9 – Validating and Launching the Instance

- After reviewing all the checkpoints, click “Launch Instance” as shown in figure 9.

- After these steps we will launch our first instance.

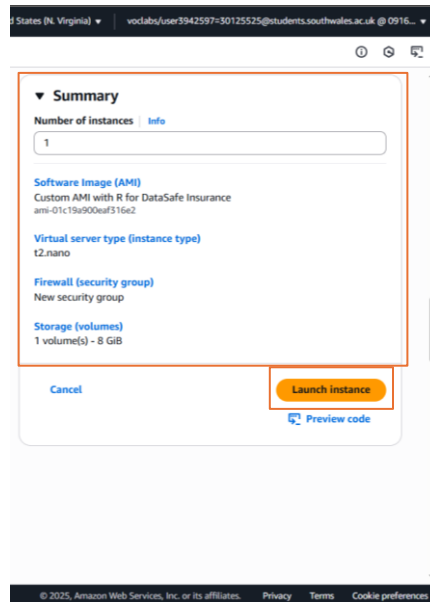


Figure 9. Launching Instance.

Step 10 – Navigating the Test Instance.

- Under “**Instances**” we can find our newly created test instance running with all checks cleared as highlighted in figure 10.

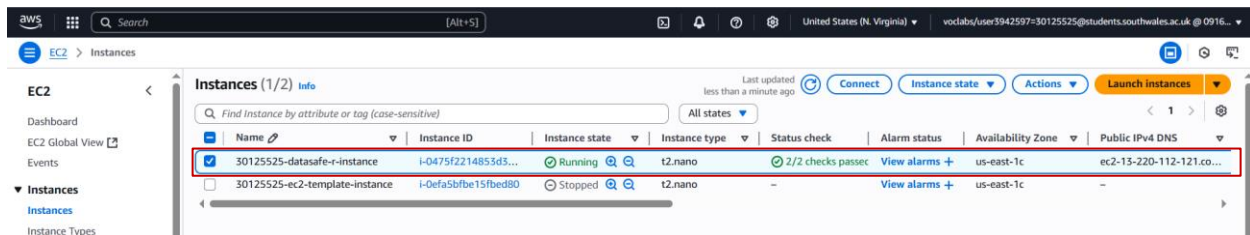


Figure 10. Navigating New Test Instance.

Step 11 – Launching the Bash Terminal to run verification commands

- Select the newly created instance and click “**Connect**” as shown in figure 11.
- Then, select “**Connect**” again as shown in figure 12.

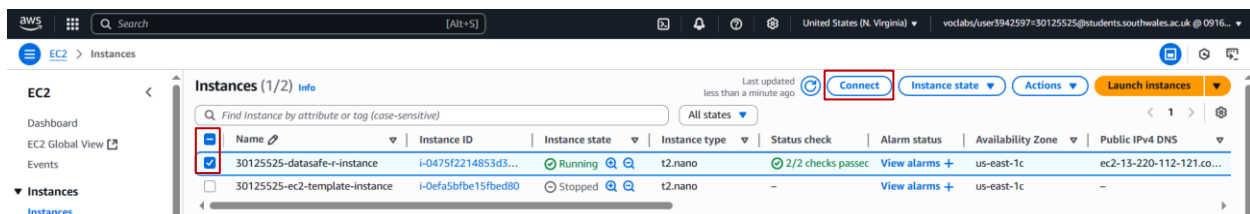


Figure 11. Connecting to Ubuntu Terminal.

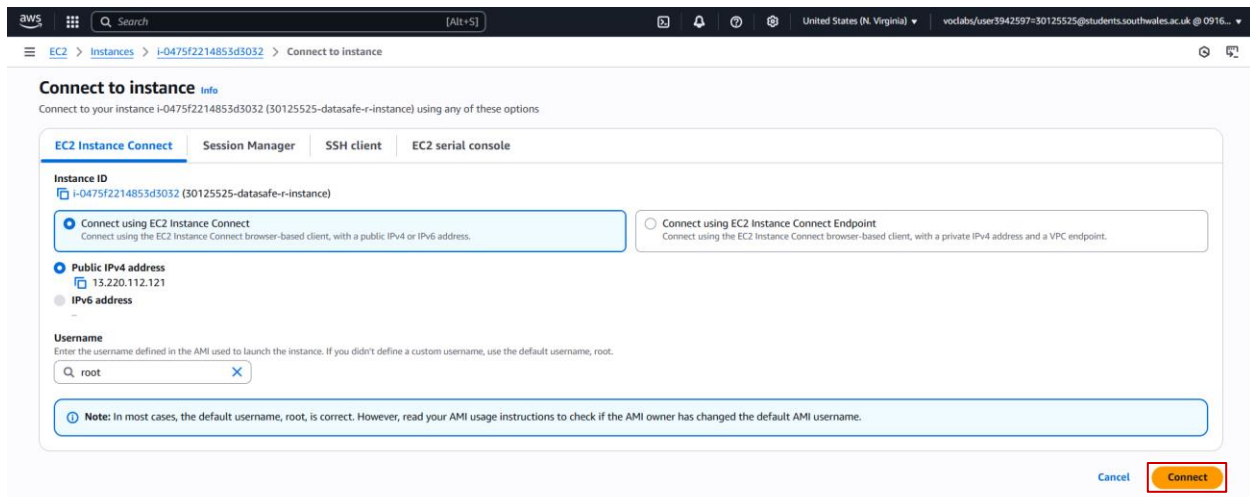


Figure 12. Last step to connect Ubuntu.

Validation Commands -

a. nano 30125525_script.R

- Creating a new R script file.

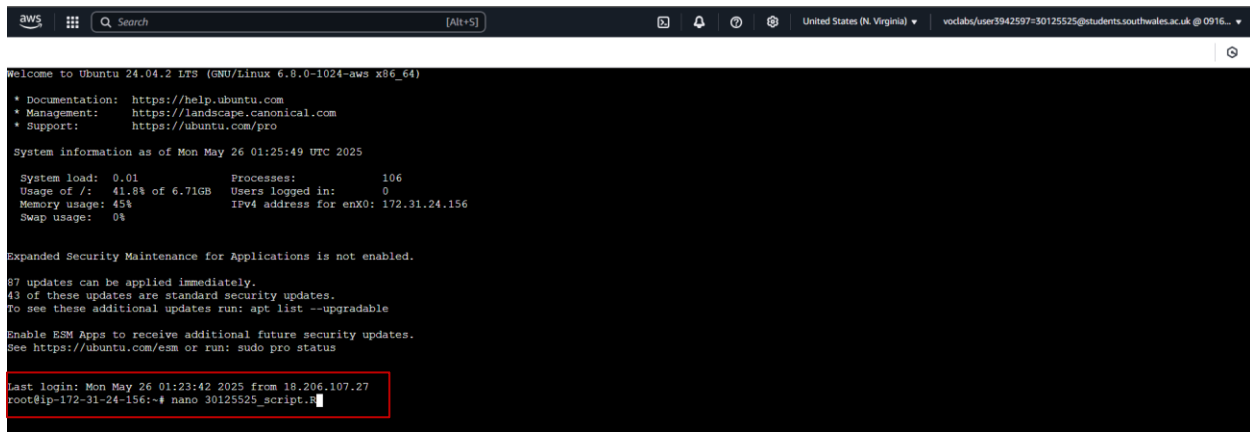


Figure 13. Creating an R File.

b. cat('30125525. R has been installed successfully for DataSafe Insurance Analytics!\n').

- Editing the file content.

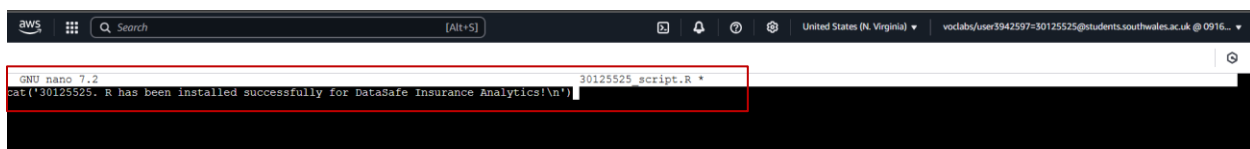


Figure 14. Editing and appending data into the R File.

c. Ctrl + S

- Saving the file.

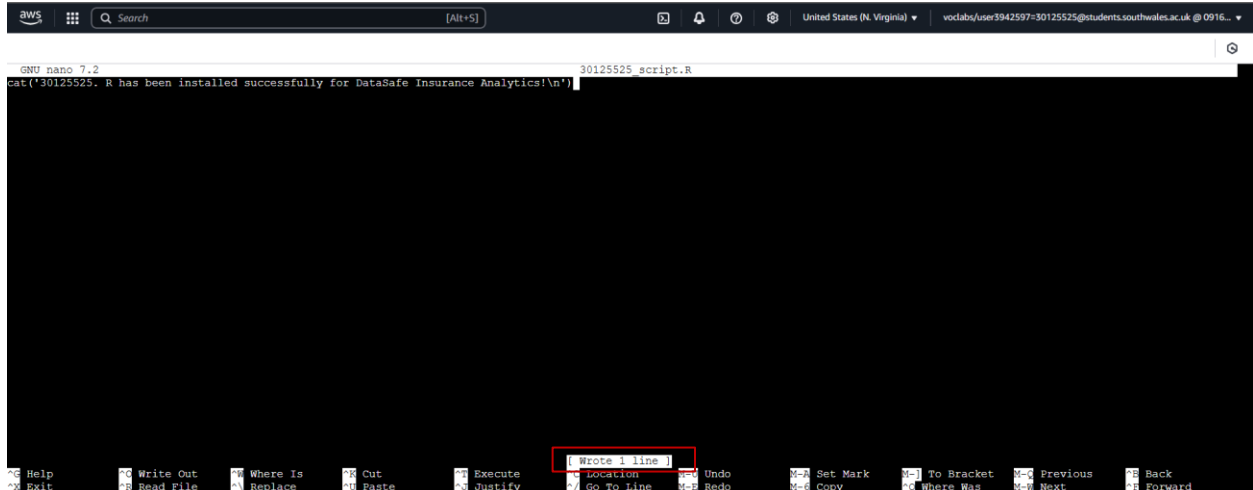


Figure 15. Saving the R File.

d. Ctrl + X

- Exiting the file.

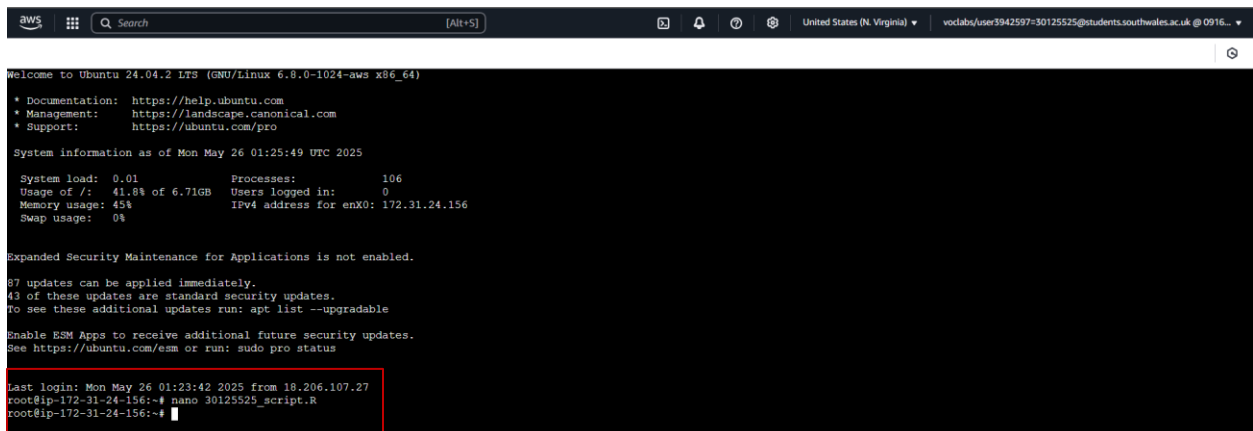


Figure 16. Exiting the R File.

e. Rscript 30125525_script.R

- Running the R script created above.

```

Welcome to Ubuntu 24.04.2 LTS (GNU/Linux 6.8.0-1024-aws x86_64)

* Documentation:  https://help.ubuntu.com
* Management:    https://landscape.canonical.com
* Support:        https://ubuntu.com/pro

System information as of Mon May 26 01:25:49 UTC 2025

System load:  0.01   Processes:           106
Usage of /:   41.8% of 6.71GB   Users logged in:   0
Memory usage: 45%    IPv4 address for enX0: 172.31.24.156
Swap usage:   0%

Expanded Security Maintenance for Applications is not enabled.

87 updates can be applied immediately.
43 of these updates are standard security updates.
To see these additional updates run: apt list --upgradable

Enable ESM Apps to receive additional future security updates.
See https://ubuntu.com/esm or run: sudo pro status

Last login: Mon May 26 01:23:42 2025 from 18.206.107.27
root@ip-172-31-24-156:~# nano 30125525_script.R
root@ip-172-31-24-156:~# Rscript 30125525_script.R
30125525. R has been installed successfully for DataSafe Insurance Analytics!
root@ip-172-31-24-156:~#
```

i-0475f2214853d3032 (30125525-datasafe-r-instance)

PublicIPs: 13.220.112.121 PrivateIPs: 172.31.24.156

Figure 17. Running the R Script.

Step 12 – Stop required running instances.

- Ensure to stop **unrequired** running instances after completing the work. Doing this will save us a lot of money.
- Select the instance you want to stop and click “**Instance State**” and the click “**Stop Instance**” as shown in figure 18.

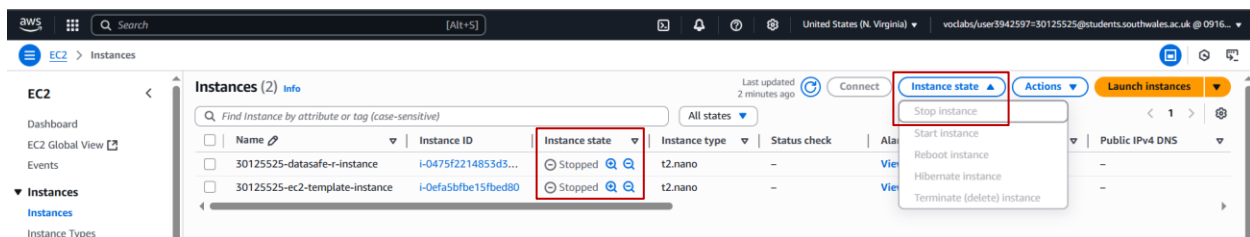


Figure 18. Stopping unwanted running instances.

References –

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